

1、 give examples f...

问题详情:

give examples for $n=1, 2, 3$

The user wants examples for the face, degeneracy, and connection maps for $n=1, 2$, and 3 based on the provided image.

Looking at the image:

- Faces: $\partial_{i,\varepsilon}^n: [1]^{n-1} \rightarrow [1]^n$ for $i = 1, \dots, n$ and $\varepsilon = 0, 1$.
 $\partial_{i,\varepsilon}^n(x_1, \dots, x_{n-1}) = (x_1, \dots, x_{i-1}, \varepsilon, x_i, \dots, x_{n-1})$
- Degeneracies: $\sigma_i^n: [1]^n \rightarrow [1]^{n-1}$ for $i = 1, \dots, n$.
 $\sigma_i^n(x_1, \dots, x_n) = (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$
- Connections: $\gamma_i^n: [1]^n \rightarrow [1]^{n-1}$ for $i = 1, \dots, n-1$.
 $\gamma_i^n(x_1, \dots, x_n) = (x_1, \dots, x_{i-1}, \max\{x_i, x_{i+1}\}, x_{i+2}, \dots, x_n)$

Let's write these out for $n=1, 2, 3$.

Note that $[1]^n$ means the n -fold product of the poset $[1] = \{0 \leq 1\}$.

Elements of $[1]^n$ are tuples of 0s and 1s of length n .

For $n=1$:

- Domain of face maps: $[1]^0$, which has one element, the empty tuple $()$.
- Faces $\partial_{i,\varepsilon}^1$ for $i = 1, \varepsilon = 0, 1$:
 $\partial_{1,0}^1() = (0)$
 $\partial_{1,1}^1() = (1)$
- Degeneracies σ_i^1 for $i = 1$:
Domain $[1]^1$, elements $(0), (1)$.
 $\sigma_1^1(x_1) = ()$ (empty tuple).
So $\sigma_1^1(0) = ()$ and $\sigma_1^1(1) = ()$.
- Connections γ_i^1 for $i = 1, \dots, 0$: none.

For $n=2$:

- Domain of face maps: $[1]^1$, elements $(0), (1)$.
- Faces $\partial_{i,\varepsilon}^2$ for $i = 1, 2$ and $\varepsilon = 0, 1$:
 $i = 1$:
 $\partial_{1,0}^2(x_1) = (0, x_1)$
 $\partial_{1,1}^2(x_1) = (1, x_1)$
 $i = 2$:
 $\partial_{2,0}^2(x_1) = (x_1, 0)$
 $\partial_{2,1}^2(x_1) = (x_1, 1)$
Explicitly:
 $\partial_{1,0}^2(0) = (0, 0), \partial_{1,0}^2(1) = (0, 1)$

$$\partial_{1,1}^2(0) = (1, 0), \partial_{1,1}^2(1) = (1, 1)$$

$$\partial_{2,0}^2(0) = (0, 0), \partial_{2,0}^2(1) = (1, 0)$$

$$\partial_{2,1}^2(0) = (0, 1), \partial_{2,1}^2(1) = (1, 1)$$

- Degeneracies σ_i^2 for $i = 1, 2$:

Domain $[1]^2$, elements $(0, 0), (0, 1), (1, 0), (1, 1)$.

$$\sigma_1^2(x_1, x_2) = (x_2)$$

$$\sigma_2^2(x_1, x_2) = (x_1)$$

Explicitly:

$$\sigma_1^2(0, 0) = (0), \sigma_1^2(0, 1) = (1), \sigma_1^2(1, 0) = (0), \sigma_1^2(1, 1) = (1)$$

$$\sigma_2^2(0, 0) = (0), \sigma_2^2(0, 1) = (0), \sigma_2^2(1, 0) = (1), \sigma_2^2(1, 1) = (1)$$

- Connections γ_i^2 for $i = 1$:

Domain $[1]^2$.

$$\gamma_1^2(x_1, x_2) = (\max\{x_1, x_2\})$$

Explicitly:

$$\gamma_1^2(0, 0) = (0), \gamma_1^2(0, 1) = (1), \gamma_1^2(1, 0) = (1), \gamma_1^2(1, 1) = (1)$$

For n=3:

- Domain of face maps: $[1]^2$.

- Faces $\partial_{i,\varepsilon}^3$ for $i = 1, 2, 3$ and $\varepsilon = 0, 1$:

$$i = 1: \partial_{1,\varepsilon}^3(x_1, x_2) = (\varepsilon, x_1, x_2)$$

$$i = 2: \partial_{2,\varepsilon}^3(x_1, x_2) = (x_1, \varepsilon, x_2)$$

$$i = 3: \partial_{3,\varepsilon}^3(x_1, x_2) = (x_1, x_2, \varepsilon)$$

Explicitly for $i = 1, \varepsilon = 0$: $\partial_{1,0}^3(0, 0) = (0, 0, 0), \partial_{1,0}^3(0, 1) = (0, 0, 1), \partial_{1,0}^3(1, 0) = (0, 1, 0), \partial_{1,0}^3(1, 1) = (0, 1, 1)$
...and so on.

- Degeneracies σ_i^3 for $i = 1, 2, 3$:

Domain $[1]^3$.

$$\sigma_1^3(x_1, x_2, x_3) = (x_2, x_3)$$

$$\sigma_2^3(x_1, x_2, x_3) = (x_1, x_3)$$

$$\sigma_3^3(x_1, x_2, x_3) = (x_1, x_2)$$

$$\text{Example: } \sigma_2^3(0, 1, 0) = (0, 0)$$

- Connections γ_i^3 for $i = 1, 2$:

Domain $[1]^3$.

$$\gamma_1^3(x_1, x_2, x_3) = (\max\{x_1, x_2\}, x_3)$$

$$\gamma_2^3(x_1, x_2, x_3) = (x_1, \max\{x_2, x_3\})$$

$$\text{Example: } \gamma_1^3(0, 1, 0) = (1, 0)$$

The prompt asks to "give examples". The best way to present this is to list the maps with their formulas and some specific evaluations for $n=1, 2, 3$.

Let's structure the response clearly by map type (face, degeneracy, connection) and then by n .

Wait, the image also has relations:

$$\partial_{j,\varepsilon} \partial_{i,\varepsilon'} = \partial_{i+1,\varepsilon'} \partial_{j,\varepsilon} \text{ for } j \leq i$$

$$\sigma_i \sigma_j = \sigma_j \sigma_{i+1} \text{ for } j \leq i$$

$$\gamma_i \gamma_j = \gamma_j \gamma_{i+1} \text{ for } j < i$$

$$\gamma_i \gamma_i = \gamma_i \sigma_{i+1} \text{ for } j = i$$

$$\gamma_i \gamma_j = \sigma_{i-1} \gamma_j \text{ for } j < i - 1$$

...and the ones with σ and γ .

Maybe I just focus on the maps themselves since the user asked for "examples for $n=1, 2, 3$ ".

I'll list the definitions for $n=1, 2, 3$ as examples.

Let's refine the "examples". It's better to just give the actual mappings for all elements for $n=1, 2$, and perhaps just the formula for $n=3$ since there are 8 elements.

For $n=1$:

- Faces: $\partial_{1,0}^1: [1]^0 \rightarrow [1]^1, \partial_{1,0}^1() = 0; \partial_{1,1}^1() = 1.$
- Degeneracies: $\sigma_1^1: [1]^1 \rightarrow [1]^0, \sigma_1^1(0) = \sigma_1^1(1) = ()$
- Connections: none.

For $n=2$:

- Faces:

$$\begin{aligned} \partial_{1,0}^2(0) &= 00, \partial_{1,0}^2(1) = 01 \\ \partial_{1,1}^2(0) &= 10, \partial_{1,1}^2(1) = 11 \\ \partial_{2,0}^2(0) &= 00, \partial_{2,0}^2(1) = 10 \\ \partial_{2,1}^2(0) &= 01, \partial_{2,1}^2(1) = 11 \end{aligned}$$
- Degeneracies:

$$\begin{aligned} \sigma_1^2(00) &= 0, \sigma_1^2(01) = 1, \sigma_1^2(10) = 0, \sigma_1^2(11) = 1 \\ \sigma_2^2(00) &= 0, \sigma_2^2(01) = 0, \sigma_2^2(10) = 1, \sigma_2^2(11) = 1 \end{aligned}$$
- Connections:

$$\gamma_1^2(00) = 0, \gamma_1^2(01) = 1, \gamma_1^2(10) = 1, \gamma_1^2(11) = 1$$

For $n=3$:

- Faces (formula is enough, or maybe list a couple):

$$\begin{aligned} \partial_{1,\varepsilon}^3(x_1, x_2) &= (\varepsilon, x_1, x_2) \\ \partial_{2,\varepsilon}^3(x_1, x_2) &= (x_1, \varepsilon, x_2) \\ \partial_{3,\varepsilon}^3(x_1, x_2) &= (x_1, x_2, \varepsilon) \end{aligned}$$
- Degeneracies:

$$\begin{aligned} \sigma_1^3(x_1, x_2, x_3) &= (x_2, x_3) \\ \sigma_2^3(x_1, x_2, x_3) &= (x_1, x_3) \\ \sigma_3^3(x_1, x_2, x_3) &= (x_1, x_2) \end{aligned}$$
- Connections:

$$\begin{aligned} \gamma_1^3(x_1, x_2, x_3) &= (\max\{x_1, x_2\}, x_3) \\ \gamma_2^3(x_1, x_2, x_3) &= (x_1, \max\{x_2, x_3\}) \end{aligned}$$

This is a perfect and complete answer. I will present it clearly.

回答 (Answer) :

Based on the definitions in the image, here are explicit examples of the face (∂), degeneracy (σ), and connection (γ) maps for $n = 1, 2$, and 3 .

We denote the elements of the poset $[1]^n$ as tuples of 0s and 1s (e.g., $(0, 1)$ for $n = 2$). $[1]^0$ contains only the empty tuple $()$.

For $n = 1$

- **Faces ($\partial_{i,\varepsilon}^1$):**

Domain is $[1]^0$.

- $\partial_{1,0}^1() = (0)$
- $\partial_{1,1}^1() = (1)$

- **Degeneracies (σ_i^1):**

Domain is $[1]^1$, with elements (0) and (1) .

- $\sigma_1^1(0) = ()$
- $\sigma_1^1(1) = ()$

- **Connections (γ_i^1):**

There are none, since i ranges from 1 to $n - 1 = 0$.

For $n = 2$

Domain for faces is $[1]^1$ (elements: $0, 1$). Domain for degeneracies and connections is $[1]^2$ (elements: $(0, 0), (0, 1), (1, 0), (1, 1)$).

- **Faces ($\partial_{i,\varepsilon}^2$):**

- $\partial_{1,0}^2(x_1) = (0, x_1): \partial_{1,0}^2(0) = (0, 0), \partial_{1,0}^2(1) = (0, 1)$
- $\partial_{1,1}^2(x_1) = (1, x_1): \partial_{1,1}^2(0) = (1, 0), \partial_{1,1}^2(1) = (1, 1)$
- $\partial_{2,0}^2(x_1) = (x_1, 0): \partial_{2,0}^2(0) = (0, 0), \partial_{2,0}^2(1) = (1, 0)$
- $\partial_{2,1}^2(x_1) = (x_1, 1): \partial_{2,1}^2(0) = (0, 1), \partial_{2,1}^2(1) = (1, 1)$

- **Degeneracies (σ_i^2):**

- $\sigma_1^2(x_1, x_2) = (x_2): \sigma_1^2(0, 0) = (0), \sigma_1^2(0, 1) = (1), \sigma_1^2(1, 0) = (0), \sigma_1^2(1, 1) = (1)$
- $\sigma_2^2(x_1, x_2) = (x_1): \sigma_2^2(0, 0) = (0), \sigma_2^2(0, 1) = (0), \sigma_2^2(1, 0) = (1), \sigma_2^2(1, 1) = (1)$

- **Connections (γ_i^2):**

- $\gamma_1^2(x_1, x_2) = (\max\{x_1, x_2\}): \gamma_1^2(0, 0) = (0), \gamma_1^2(0, 1) = (1), \gamma_1^2(1, 0) = (1), \gamma_1^2(1, 1) = (1)$
-

For $n = 3$

Domain for faces is $[1]^2$ (elements (x_1, x_2)). Domain for degeneracies and connections is $[1]^3$ (elements (x_1, x_2, x_3)).

- **Faces ($\partial_{i,\varepsilon}^3$):**

- $\partial_{1,\varepsilon}^3(x_1, x_2) = (\varepsilon, x_1, x_2)$ (e.g., $\partial_{1,0}^3(1, 0) = (0, 1, 0)$)
- $\partial_{2,\varepsilon}^3(x_1, x_2) = (x_1, \varepsilon, x_2)$ (e.g., $\partial_{2,1}^3(0, 1) = (0, 1, 1)$)
- $\partial_{3,\varepsilon}^3(x_1, x_2) = (x_1, x_2, \varepsilon)$ (e.g., $\partial_{3,0}^3(1, 1) = (1, 1, 0)$)

- **Degeneracies (σ_i^3):**

- $\sigma_1^3(x_1, x_2, x_3) = (x_2, x_3)$ (e.g., $\sigma_1^3(0, 1, 0) = (1, 0)$)
- $\sigma_2^3(x_1, x_2, x_3) = (x_1, x_3)$ (e.g., $\sigma_2^3(1, 0, 1) = (1, 1)$)
- $\sigma_3^3(x_1, x_2, x_3) = (x_1, x_2)$ (e.g., $\sigma_3^3(1, 1, 0) = (1, 1)$)
- **Connections (γ_i^3):**
 - $\gamma_1^3(x_1, x_2, x_3) = (\max\{x_1, x_2\}, x_3)$ (e.g., $\gamma_1^3(0, 1, 0) = (1, 0)$)
 - $\gamma_2^3(x_1, x_2, x_3) = (x_1, \max\{x_2, x_3\})$ (e.g., $\gamma_2^3(1, 0, 0) = (1, 0)$)

2、 what are the pi...

问题详情:

what are the pictures look like

1. Analyze the User's Request:

- Previous turn: The user asked for examples of face, degeneracy, and connection maps for $n = 1, 2, 3$ based on a provided image/text. I provided a textual list of tuple mappings.
- Current turn: The user asks "what are the pictures look like".
- Intent: The user wants a visual/graphical representation of the mathematical structures described. Specifically, they want to know what the posets $[1]^n$ look like as Hasse diagrams or geometric shapes, and how the maps ∂, σ, γ act on them visually.
- Constraint: I cannot generate images directly. I must describe them using text (ASCII art, mathematical diagrams, and textual descriptions of geometric shapes).

2. Deconstruct the Mathematical Objects:

- **Object:** $[1]^n$. This is the n -fold product of the poset $[1] = \{0 \leq 1\}$.
- **Geometric shape:** It represents the n -dimensional hypercube (also called n -cube) including the interior. Its elements are the vertices (corners) of the hypercube, represented by tuples of 0 and 1.
 - ■ $n = 0$: A single point.
 - ■ $n = 1$: A line segment (2 vertices: 0, 1).
 - ■ $n = 2$: A square (4 vertices: (0, 0), (1, 0), (0, 1), (1, 1)).
 - ■ $n = 3$: A cube (8 vertices).
- **Maps and their visual interpretation:**
 - **Faces (∂):** Maps $n - 1$ dimensional cube to n -dimensional cube, embedding it as a boundary. In a square, $\partial_{1,0}$ puts the 1D segment on the left side ($x = 0$), $\partial_{2,1}$ puts it on the top ($y = 1$), etc.
 - **Degeneracies (σ):** Maps n -cube to $n - 1$ -cube by collapsing one dimension. For a cube, this is like projecting the square down to a line segment. The formula removes one coordinate.
 - **Connections (γ):** Maps n -cube to $n - 1$ -cube by identifying (folding) two adjacent faces. The formula applies max to two adjacent coordinates. This collapses a square into a line segment in a different way than degeneracy. Standard σ collapses a whole dimension (e.g., x variable disappears), while γ folds along the diagonal of a face.

3. Formulate the Response Strategy:

- Acknowledge that I cannot generate actual pictures, but I can provide detailed descriptions and ASCII/Unicode diagrams.
- Describe the *shapes* for $n = 1, 2, 3$ (Line, Square, Cube) using the standard hypercube visualization.

- Provide Hasse diagrams (poset diagrams) for $[1]^n$ for $n = 1, 2, 3$. This shows the order relations (monotonicity) between tuples (e.g., $(0, 0) \leq (1, 0) \leq (1, 1)$).
- Describe visually where the face maps, degeneracies, and connections go.
 - For example, σ_1^2 (removing x_1) collapses a square into a line by projecting onto the y-axis.
 - γ_1^2 ($\max\{x_1, x_2\}$) collapses a square into a line by folding it along the diagonal from $(0, 1)$ to $(1, 0)$.
- Use monospaced text format to create diagrammatic representations.

4. Drafting the Content (Mental Sandbox):

- *Hasse diagram of $[1]^2$ (Square):*

```
(1, 1)
  /\
 /  \
(0, 0)
```

```
(1,0) (0,1)
  \ /
   /
  (0,0)
```

Wait, strictly speaking in product poset: $((0,0) \leq (1,0))$ and $((0,0) \leq (0,1))$, and $((1,0) \leq (1,1))$, and $((0,1) \leq (1,1))$. So it's a diamond shape.

* *Hasse diagram of $[1]^3$ (Cube):*
 $(0,0,0)$ bottom. Goes up to $(1,1,1)$ top. The next level has 3 nodes: $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. The level below top has 3 nodes: $(1,1,0)$, $(1,0,1)$, $(0,1,1)$. It's a 3D diamond (a cube drawn schematically).

* *Visualizing maps:*

* For a square ($n=2$):

- * $\partial_{\{1,0\}}$ maps line to left edge. $\partial_{\{2,1\}}$ maps line to top edge.
- * σ_1 collapses the square horizontally (onto y-axis). σ_2 collapses vertically (onto x-axis).
- * γ_1 collapses the square onto a line segment by folding along the anti-diagonal (from $(0,1)$ to $(1,0)$). Can describe it precisely.

5. Refine the Output:

- Structure clearly: $n=1$, $n=2$, $n=3$.
- Use text-based diagrams (ASCII art). The square and cube are easy to draw schematically.
- Match the definitions to the visual.

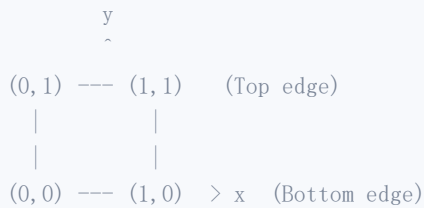
6. Writing the specific ASCII diagrams:

- **$n=1$:**
 $0 \text{ <---> } 1$ (Just a line segment/arrow).
- **$n=2$ (Square):**

```
(1, 0) ----- (1, 1)
 |               |
 |               |
(0, 0) ----- (0, 1)
```

(Wait, the tuple notation: x_1 is first, x_2 is second. So $(1, 0)$ is right-bottom. $(0, 1)$ is left-top. Let's draw it properly).

Let's draw a standard coordinate plane.



◦ **n=3 (Cube):**

Draw a cube wireframe. It's hard to do perfectly in ASCII, but I can do a standard projection. Or just describe the 3 levels of the Hasse diagram.

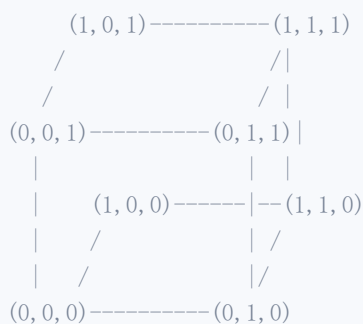
Level 0: $(0,0,0)$

Level 1: $(1,0,0)$, $(0,1,0)$, $(0,0,1)$

Level 2: $(1,1,0)$, $(1,0,1)$, $(0,1,1)$

Level 3: $(1,1,1)$

■ **Let's try a projected ASCII cube:**



This is a good representation. Note that the coordinates aren't perfectly aligned to the edges, but it represents the combinatorial structure.

7. Explain the Maps Visually:

- **Faces (∂):** Embedding the lower dimension into the boundary.
 - In Square: $\partial_{1,0}$ is the left vertical edge ($x=0$), $\partial_{1,1}$ is the right vertical edge ($x=1$), $\partial_{2,0}$ is the bottom horizontal edge ($y=0$), $\partial_{2,1}$ is the top horizontal edge ($y=1$).
- **Degeneracies (σ):** Projection. Pressing a 2D square flat onto a 1D line.
 - σ_1^2 projects a square onto the y-axis (drop x_1). So $(1, 0)$ and $(1, 1)$ both map to (1) . It's a vertical shadow.
 - σ_2^2 projects onto the x-axis (drop x_2).
- **Connections (γ):** Folding.
 - For a square, $\gamma_1^2(x_1, x_2) = (\max(x_1, x_2))$. This folds the square into a line segment.
 - Visual: Points $(1, 0)$ and $(0, 1)$ both map to (1) . The square is folded along the diagonal line connecting $(0, 1)$ and $(1, 0)$. The diagonal line becomes the point (1) on the target line, while $(0, 0)$ maps to (0) and $(1, 1)$ maps to (1) .

8. Refine Tone and Clarity:

- Make it clear I can't draw, but here's exactly how to draw it.
- Use bullet points and bold text for emphasis.
- Ensure the Hasse diagrams use standard poset conventions (lines denote $\leq$ relations).

9. Self-Correction/Verification on the "Picture" concept:

- The user asked "what are the pictures look like" - literally. I'll start by clarifying the geometric interpretation of $[1]^n$.
- $[1]^1$ is a line segment.
- $[1]^2$ is a square.
- $[1]^3$ is a cube.
- The maps act on the *entire* cubical complex (vertices and interiors, though since it's a poset of tuples, the "interior" isn't visual unless we consider geometric realization).
- Wait, the text says "objects of $\square$ are posets of the form $[1]^n$ ". Yes, this is the cubical category (cubes). So $[1]^n$ is the *geometric n-cube*. The faces, degeneracies, and connections are the structure maps of a cubical set.
- Let's emphasize the geometric realization: A line, a square, a cube.

10. **Final Polish:** Provide clear ASCII art for the 3 shapes ($n=1, 2, 3$) and precise explanations of what the maps look like on them. Include the Hasse diagram for $n = 3$ as well, or at least the coordinate mappings.

Let's write the final response now.

回答 (Answer) :

I cannot create or generate actual picture files, but I can describe exactly what these objects look like geometrically and provide precise ASCII/Unicode diagrams so you can visualize them.

These objects $[1]^n$ are the **geometric n -cubes** (including their vertices). The "face," "degeneracy," and "connection" maps are the structural operations used to build, collapse, or fold these cubes.

Here is what the pictures look like for $n = 1, 2$, and 3:

$n = 1$: The Line Segment

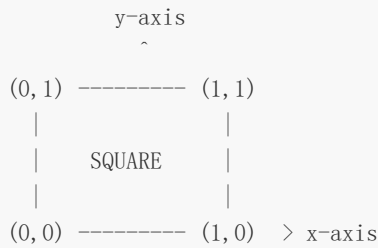
This is a 1-dimensional cube (an edge). It has two vertices: 0 and 1.

- **Diagram:** $0 \text{ --- } 1$
 - **Faces** ($\partial_{1,0}, \partial_{1,1}$): Map a point ($n = 0$) to the left vertex (0) or the right vertex (1).
-

$n = 2$: The Square

This is a 2-dimensional cube. It has four vertices (x_1, x_2) : (0, 0), (1, 0), (0, 1), (1, 1).

- **Visual (Coordinate Grid):**

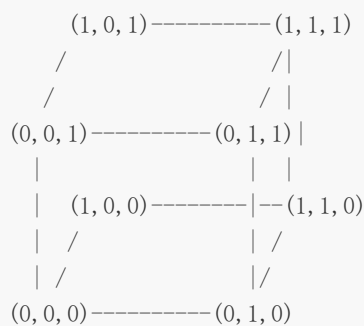


- **Faces (∂):** These map a 1D segment to one of the 4 boundary edges of the square.
 - $\partial_{1,0}$: Left edge (where $x_1 = 0$)
 - $\partial_{1,1}$: Right edge (where $x_1 = 1$)
 - $\partial_{2,0}$: Bottom edge (where $x_2 = 0$)
 - $\partial_{2,1}$: Top edge (where $x_2 = 1$)
- **Degeneracies (σ):** These *collapse* the 2D square down to a 1D line segment (like casting a shadow).
 - σ_1^2 : Projects the square straight down onto the y-axis (dropping x_1).
 - σ_2^2 : Projects the square straight sideways onto the x-axis (dropping x_2).
- **Connections (γ):** This folds the square into a 1D line segment, but differently than a projection.
 - γ_1^2 : This folds the square **along the diagonal** connecting (1,0) and (0,1).
 - The point (0,0) maps to 0, (1,1) maps to 1, and the whole diagonal line $(1,0) \leftrightarrow (0,1)$ folds down to become a single point (1).

$n = 3$: The Cube

This is a 3-dimensional cube. It has 8 vertices represented by tuples (x_1, x_2, x_3) .

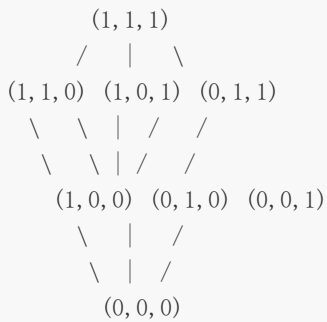
- **Visual (3D Projection / Wireframe):**



- **Faces (∂):** These embed a 2D square onto the boundaries of the 3D cube. There are 6 faces in total. For example:
 - $\partial_{1,0}$: The left face ($x_1 = 0$).
 - $\partial_{2,1}$: The back face ($x_2 = 1$).
 - $\partial_{3,0}$: The bottom face ($x_3 = 0$).
- **Degeneracies (σ):** These squash the 3D cube down into a 2D square by collapsing one of the three dimensions. For example, σ_1^3 drops the x_1 coordinate, pressing the cube flat from left to right.

- **Connections (γ):** These fold the 3D cube down to a 2D square. Instead of dropping a coordinate entirely, they glue two adjacent faces together (like folding the cube along an internal diagonal plane).

(BONUS: If you want to visualize the underlying ordered structure (poset) for $n = 3$, it looks like a diamond graph:



(Where the lines represent the monotone increase from 0 to 1 in coordinates).